

Prob. $f(t) = \cos 3t$

$$f'(t) = -3 \sin 3t$$

$$\mathcal{L}\{-3 \sin 3t\} = s \mathcal{L}\{\cos 3t\} - 1$$

$$\mathcal{L}\{\cos 3t\} = \frac{\frac{-9}{s^2+9} + 1}{s} = \frac{s}{s^2+9} \quad \text{olur.}$$

Prob.

$$f(t) = \begin{cases} -1 & , t \leq 4 \\ 1 & , t > 4 \end{cases} \quad \mathcal{L}\{f\} = ?$$

$$\begin{aligned} \mathcal{L}\{f\} &= \int_0^{\infty} e^{-st} f(t) dt = \int_0^4 -e^{-st} dt + \int_4^{\infty} e^{-st} dt \\ &= \frac{2e^{-4s}}{s} - \frac{1}{s} \quad (s > 0) \end{aligned}$$

Prob.

$$f(t) = \begin{cases} 5 & , 0 < t < 3 \\ 0 & , t > 3 \end{cases}$$

Prob. $\mathcal{L}\{\sinh at\} = \mathcal{L}\left\{ \frac{e^{at} - e^{-at}}{2} \right\}$

$$= \frac{1}{2} \cdot \frac{1}{s-a} - \frac{1}{2} \cdot \frac{1}{s+a} = \frac{a}{s^2 - a^2}$$

Ödev: $\mathcal{L}\{\cos at\}$

Ödev: $\mathcal{L}\{\cos^2 at\}$

Özellik III: $t > 0$ olmak üzere $f(t)$ fonksiyonunun Laplace dönüşümü var olsun.

0 zaman $t > 0$ için

$\lim_{t \rightarrow \infty} f(t)e^{-st} = 0$ olmak üzere

$$\mathcal{L}\{f^{(n)}(t)\} = s^n \mathcal{L}\{f(t)\} - s^{n-1} f(0) - s^{n-2} f'(0) - s^{n-3} f''(0) - \dots - f^{(n-1)}(0)$$

ÖRNEK $f(t) = \sin bt$ için $\mathcal{L}\{f''(t)\} = ?$

$$s^2 \mathcal{L}\{f(t)\} - s f(0) - f'(0) = \mathcal{L}\{f''(t)\}$$

$$= s^2 \frac{b}{s^2 + b^2} - b$$

$$f'(t) = b \cos bt$$

Özellik IV $s > a$ için $\mathcal{L}\{f\}$ var olacak biçimde bir f fonksiyonu verilsin. O zaman her a sabiti için

$$F(s) = \mathcal{L}\{f(t)\} \text{ ise}$$

$$\mathcal{L}\{e^{at} f(t)\} = F(s-a) \quad (s > a)$$

ÖRNEK $\mathcal{L}\{e^{at} t\} = ?$

GÖZÜM

$$f(t) = t \quad \mathcal{L}\{f(t)\} = \mathcal{L}\{t\} = \frac{1}{s^2} = F(s)$$

$$\text{O halde } \mathcal{L}\{e^{at} t\} = F(s-a) = \frac{1}{(s-a)^2}, \quad s > a$$

ÖRNEK $\mathcal{L}\{e^{-t} \cos 2t\}$

GÖZÜM

$$a = -1 \quad f(t) = \cos 2t$$

$$F(s) = \mathcal{L}\{\cos 2t\} = \frac{s}{s^2 + 4}$$

$$\mathcal{L}\{e^{-t} \cos 2t\} = F(s-a) = F(s+1) = \frac{s-a}{(s-a)^2 + 4}$$

$$= \frac{s+1}{(s+1)^2 + 4} \quad (s > -1)$$

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Ödev : $\mathcal{L}\{e^{at} \sin bt\}$

Problem :

a) $\mathcal{L}\{e^{3t} + t^2\}$

b) $\mathcal{L}\{e^{2t} \sinh 3t\}$

Özellik V $\mathcal{L}\{f(t)\} = F(s)$ olsun. O zaman

a) $\mathcal{L}\{t^n f(t)\} = (-1)^n \frac{d^n}{ds^n} [F(s)] = (-1)^n F^{(n)}(s)$
 $n=1, 2, 3, \dots$

b) $\mathcal{L}\{f(at)\} = \frac{1}{a} F\left(\frac{s}{a}\right)$

ÖRNEK $\mathcal{L}\{\sin 3t\} = \frac{1}{3} \cdot \frac{1}{\frac{s^2}{9} + 1} = \frac{3}{s^2 + 9}$

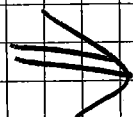
$$F(s) = \mathcal{L}\{\sin t\} = \frac{1}{s^2 + 1}$$

ÖRNEK $\mathcal{L}\{t^2 \sin bt\}$

Çözüm

$$f(t) = \sin bt \quad F(s) = \mathcal{L}\{\sin bt\} = \frac{b}{s^2 + b^2}$$

$$F'(s) = \frac{-2sb}{(s^2 + b^2)^2}$$



$$\Rightarrow F^{(2)}(s) = \frac{6bs^2 - 2b^3}{(s^2 + b^2)^3}$$

Ödev: $\mathcal{L}\{t^2 \sin bt\} = (-1)^2 \quad F^{(2)}(s) = ?$

ÖDEV: a) $\mathcal{L}\{te^{2t}\}$

b) $\mathcal{L}\{t^2 e^{2t}\}$

c) $\mathcal{L}\{t^2 e^{4t}\} =$

$$f(t) = e^{4t} \quad \mathcal{L}\{f(t)\} = F(s) = \frac{1}{s-4}$$

$$F^{(2)}(s) = \frac{2}{(s-4)^3}$$

Ters Dönüşüm :

Bu bölümde bir önceki bölümün tam tersi yapılır. Laplace dönüşümü verilecek buradan yola çıkarak fonksiyon bulmak hedeflenecek.

$$f(t) = \mathcal{L}^{-1}\{F(s)\}$$

Buna F 'nin ters dönüşümü adı verilir.

Burada $\mathcal{L}\{f(t)\} = F(s)$ dir.

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Teorem : f ve g , $t \geq 0$ için sürekli fonksiyonlar olsunlar. f ve g 'nin Laplace dön. aynı ise $\forall t \geq 0$ için $f(t) = g(t)$ 'dir

Ters dönüşümleri bulmak için aşağıdaki tablo kullanılır.

$$F(s) = \mathcal{L}\{f(t)\}$$

$$f(t) = \mathcal{L}^{-1}\{F(s)\}$$

$$\frac{1}{s}$$

$$1$$

$$\frac{1}{s^2}$$

$$t$$

$$\frac{1}{s-a}$$

$$e^{at}$$

$$\frac{a}{s^2+a^2} \quad (s>0)$$

$$\sin at$$

$$\frac{s}{s^2+a^2} \quad (s>0)$$

$$\cos at$$

$$\frac{a}{s^2-a^2} \quad (s>|a|)$$

$$\sinh at$$

$$\frac{s}{s^2-a^2} \quad (s>|a|)$$

$$\cosh at$$

ÖRNEK

$$1) \mathcal{L}^{-1} \left\{ \frac{2}{s^3} \right\} = t^2$$

$$2) \mathcal{L}^{-1} \left\{ \frac{s}{s^2 + 25} \right\} = \cos 5t$$

$$3) \mathcal{L}^{-1} \left\{ \frac{s}{s^2 - 25} \right\} = \cosh 5t$$

$$4) \mathcal{L}^{-1} \left\{ \frac{2s}{(s^2 + 1)^2} \right\} = t \sin t$$

$$5) \mathcal{L}^{-1} \left\{ \frac{1}{s+5} \right\} = e^{-5t}$$

Prob. Tabloyu kullanarak

$$\mathcal{L}^{-1} \left\{ \frac{1}{s^2 + 6s + 13} \right\}$$

bulunuz.

$$= \mathcal{L}^{-1} \left\{ \frac{1}{(s+3)^2 + 4} \right\} = \mathcal{L}^{-1} \left\{ \frac{1}{2} \cdot \frac{2}{(s+3)^2 + 4} \right\}$$

$$= \frac{1}{2} \mathcal{L}^{-1} \left\{ \frac{2}{(s+3)^2 + 4} \right\} = \frac{1}{2} e^{-3t} \sin 2t$$

$$\mathcal{L} \{ e^{bt} \sin at \} = \frac{a}{(s-b)^2 + a^2} \quad s > b$$

Sabit Katsayılı Doğrusal Diferansiyel Denklemlerin

Laplace Dönüşüm Çözümleri

$$a_0 \frac{d^n y}{dt^n} + a_1 \frac{d^{n-1} y}{dt^{n-1}} + \dots + a_{n-1} \frac{dy}{dt} + a_n y = G(t)$$

$$y(0) = c_0, \quad y'(0) = c_1, \quad \dots, \quad y^{(n-1)}(0) = c_{n-1}$$

başlangıç değer problemi

I- Verilen denklemin her iki tarafına Laplace dönüşümü uygulanır.

$$\mathcal{L}\left\{\frac{d^n y}{dt^n}\right\} = s^n \mathcal{L}\{y(t)\} - s^{n-1} y(0) - s^{n-2} y'(0) - \dots - y^{(n-1)}(0)$$

eşitliği uygulanarak denklem düzenlenir.

III- $\mathcal{L}\{y(t)\}$ getirilir ve $y = \mathcal{L}^{-1}\{\mathcal{L}\{y(t)\}\}$ elde edilir.

Bu şekilde bir tek çözüm vardır.

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ÖRNEK

$$\begin{aligned} \frac{dy}{dt} - 2y &= e^{5t} \\ y(0) &= 3 \end{aligned}$$

Boulapic doper
problemini
cözüñüz.

Gözüm

Denklemin her iki tarafına laplace dön. uypulayalım.

$$\mathcal{L}\left\{\frac{dy}{dt} - 2y\right\} = \mathcal{L}\{e^{5t}\}$$

$$\mathcal{L}\left\{\frac{dy}{dt}\right\} - 2\mathcal{L}\{y(t)\} = \frac{1}{s-5}$$

↓
s. $\mathcal{L}\{y(t)\} - y(0)$ yerine yazalım

$$s\mathcal{L}\{y(t)\} - 3 - 2\mathcal{L}\{y(t)\} = \frac{1}{s-5}$$

$$\mathcal{L}\{y(t)\}(s-2) - 3 = \frac{1}{s-5}$$

$$\mathcal{L}\{y(t)\} = \frac{3s-14}{(s-2)(s-5)} \Rightarrow \text{ters laplace dön. uypulanır}$$

$$y(t) = \mathcal{L}^{-1}\left\{\frac{3s-14}{(s-2)(s-5)}\right\}$$

Barit kevirilere girmo
yön. ile



$$y(t) = \mathcal{L}^{-1} \left\{ \frac{8}{3} \cdot \frac{1}{(s-2)} + \frac{1}{3} \cdot \frac{1}{(s-5)} \right\}$$

$$y(t) = \frac{8}{3} e^{2t} + \frac{1}{3} e^{5t}$$

ÖRNEK $\frac{d^2 y}{dt^2} - 2 \frac{dy}{dt} - 8y = 0$

$$y(0) = 3, \quad y'(0) = 6$$

GÖZÜM

$$\mathcal{L} \left\{ \frac{d^2 y}{dt^2} \right\} - 2 \mathcal{L} \left\{ \frac{dy}{dt} \right\} - 8 \mathcal{L} \{ y(t) \} = 0$$

I

II

$$\text{I: } s^2 \mathcal{L} \{ y(t) \} - s \cdot 3 - 6$$

$$\text{II: } s \cdot \mathcal{L} \{ y(t) \} - 3$$

$$s^2 \mathcal{L} \{ y(t) \} - 3s - 6 - 2s \mathcal{L} \{ y(t) \} + 6 - 8 \mathcal{L} \{ y(t) \} = 0$$

$$\mathcal{L} \{ y(t) \} (s^2 - 2s - 8) - 3s = 0$$

$$\mathcal{L} \{ y(t) \} = \frac{3s}{(s-4)(s+2)}$$



$$y(t) = \mathcal{L}^{-1} \left\{ \frac{3s}{(s-4)(s+2)} \right\}$$

$$y(t) = \mathcal{L}^{-1} \left\{ \frac{2}{s-4} + \frac{1}{s+2} \right\}$$

$$y(t) = 2 \cdot e^{4t} + e^{-2t}$$

$$\mathcal{L}\{\cos bt\} = \frac{s}{s^2 + b^2}$$

$$\mathcal{L}\{\sin bt\} = \frac{b}{s^2 + b^2}$$

$$\mathcal{L}\{e^{at} \sin bt\} = \frac{b}{(s-a)^2 + b^2}$$

$$\mathcal{L}\{e^{at} \cos bt\} = \frac{s-a}{(s-a)^2 + b^2}$$

! $\mathcal{L}\{f(t)\} = F(s)$ olmak üzere

$$\mathcal{L}\{e^{at} f(t)\} = F(s-a) \text{ dir.}$$

ÖRNEK

$$\frac{d^2 y}{dt^2} + y = e^{-2t} \cdot \sin t$$

$$y(0) = 0 \quad y'(0) = 0$$

ÇÖZÜM

$$s^2 \mathcal{L}\{y(t)\} + \mathcal{L}\{y(t)\} = \frac{1}{(s+2)^2 + 1}$$

$$\mathcal{L}\{y(t)\} (s^2 + 1) = \frac{1}{(s+2)^2 + 1}$$

$$y(t) = \mathcal{L}^{-1} \left\{ \frac{1}{(s^2 + 1)[(s+2)^2 + 1]} \right\}$$

$$\frac{As+B}{s^2+1} + \frac{Cs+D}{(s+2)^2+1} \Rightarrow A = -\frac{1}{8} \quad B = \frac{1}{8} \quad C = \frac{1}{8} \quad D = \frac{3}{8} \quad ?$$

$$y(t) = \mathcal{L}^{-1} \left\{ \frac{1}{(s^2+1)(s^2+4s+5)} \right\}$$

$$= -\frac{1}{8} \mathcal{L}^{-1} \left\{ \frac{s-1}{s^2+1} \right\} + \frac{1}{8} \mathcal{L}^{-1} \left\{ \frac{s+3}{s^2+4s+5} \right\}$$



$$y(t) = \frac{-1}{8} \mathcal{L}^{-1} \left\{ \frac{s}{s^2+1} \right\} + \frac{1}{8} \mathcal{L}^{-1} \left\{ \frac{1}{s^2+1} \right\} \\ + \frac{1}{8} \mathcal{L}^{-1} \left\{ \frac{s+2}{(s+2)^2+1} \right\} + \frac{1}{8} \mathcal{L}^{-1} \left\{ \frac{1}{(s+2)^2+1} \right\}$$

$$y(t) = \frac{-1}{8} \cos t + \frac{1}{8} \sin t + \frac{1}{8} e^{-2t} \cos t + \frac{1}{8} e^{-2t} \sin t$$

ÖRNEK $\frac{d^3y}{dt^3} + 4\frac{d^2y}{dt^2} + 5\frac{dy}{dt} + 2y = 10\cos t$

$$y(0)=0 \quad y'(0)=0 \quad y''(0)=3$$

Gözüm

$$\mathcal{L} \left\{ \frac{d^3y}{dt^3} \right\} = s^3 \mathcal{L} \{ y(t) \} - s^2 \cdot 0 - s \cdot 0 - 3$$

$$s^3 \mathcal{L} \{ y(t) \} - 3 + 4s^2 \mathcal{L} \{ y(t) \} + 5s \mathcal{L} \{ y(t) \} \\ + 2 \mathcal{L} \{ y(t) \} = 10 \cdot \frac{s}{s^2+1}$$

$$\mathcal{L} \{ y(t) \} (s^3 + 4s^2 + 5s + 2) = \frac{10s}{s^2+1} + 3$$



$$y(t) = \mathcal{L}^{-1}\{y(s)\} = \mathcal{L}^{-1}\left\{\frac{3s^2 + 10s + 3}{(s^2+1)(s+1)^2(s+2)}\right\}$$

$$\frac{As+B}{s^2+1} + \frac{C_1}{s+1} + \frac{C_2}{(s+1)^2} + \frac{D}{s+2}$$

$$A = -1 \quad B = 2 \quad C_1 = 2 \quad C_2 = -2 \quad D = -1$$

$$y(t) = \mathcal{L}^{-1}\left\{\frac{-s+2}{s^2+1} + \frac{2}{s+1} - \frac{2}{(s+1)^2} - \frac{1}{s+2}\right\}$$

$$= \mathcal{L}^{-1}\left\{\frac{s}{s^2+1}\right\} + 2\mathcal{L}^{-1}\left\{\frac{1}{s+1}\right\} + 2\mathcal{L}^{-1}\left\{\frac{1}{s+1}\right\}$$

$$- 2\mathcal{L}^{-1}\left\{\frac{1}{(s+1)^2}\right\} - \mathcal{L}^{-1}\left\{\frac{1}{s+2}\right\}$$

$$y(t) = -\cos t + 2\sin t + 2e^{-t} - 2te^{-t} - e^{-2t}$$

Problem $\frac{d^3 y}{dt^3} - 3\frac{d^2 y}{dt^2} + 3\frac{dy}{dt} - y = t^2 e^t$

$$y(0) = 1 \quad y'(0) = 0 \quad y''(0) = -2$$

Lösung

$$s^3 \mathcal{L}\{y(t)\} - s^2 + 2 - 3[s^2 \mathcal{L}\{y(t)\} - s] + 3s \mathcal{L}\{y(t)\} - \mathcal{L}\{y(t)\} = \frac{2}{(s-1)^3}$$



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$$\mathcal{L}\{y(t)\} (\quad) - s^2 + 2 + 3s = \frac{2}{(s-1)^3}$$

$$\mathcal{L}\{y(t)\} = \frac{s^2 - 3s + 1}{(s-1)^3} + \frac{2}{(s-1)^6} \cdot \frac{5!}{5!}$$

$$= \frac{(s-1)^2 - (s-1) - 1}{(s-1)^3} + \frac{2 \cdot 5!}{(s-1)^6 \cdot 5!}$$

$$= \frac{1}{s-1} - \frac{1}{(s-1)^2} - \frac{1}{(s-1)^3} + \frac{2 \cdot 5!}{(s-1)^6 \cdot 5!}$$

$$= \frac{1}{s-1} - \frac{1}{(s-1)^2} - \frac{2!}{(s-1)^3 \cdot 2!} + \frac{2 \cdot 5!}{(s-1)^6 \cdot 5!}$$

$$y(t) = e^t - t \cdot e^t - \frac{t^2 e^t}{2!} + \frac{2 + 5t^4}{5!} e^t$$

ÖRNEK

$$y'' + 2y' + 5y = e^{-t} \sin t$$

$$y(0) = 0 \quad y'(0) = 1$$

Çözüm

$$s^2 \mathcal{L}\{y(t)\} - 1 + 2s \mathcal{L}\{y(t)\} + 5 \mathcal{L}\{y(t)\} = \mathcal{L}\{e^{-t} \sin t\}$$

$$= \mathcal{L}\{e^{-t} \sin t\}$$

$$\mathcal{L}\{y(t)\} (s^2 + 2s + 5) - 1 = \frac{s+1}{(s+1)^2 + 1}$$

$$\mathcal{L}\{y(t)\} = \frac{s^2 + 2s + 3}{(s^2 + 2s + 5)(s^2 + 2s + 2)}$$

$$\frac{As+B}{s^2+2s+2} + \frac{Cs+D}{s^2+2s+5} \quad \text{den} \quad A=0 \quad B=\frac{1}{3}$$

$$C=0 \quad D=\frac{2}{3}$$

$$y(t) = \mathcal{L}^{-1} \left\{ \frac{\frac{1}{3}}{s^2+2s+2} + \frac{\frac{2}{3}}{s^2+2s+5} \right\}$$

$$= \frac{1}{3} \mathcal{L}^{-1} \left\{ \frac{1}{s^2+2s+2} \right\} + \frac{2}{3} \mathcal{L}^{-1} \left\{ \frac{1}{s^2+2s+5} \right\}$$

$$= \frac{1}{3} \mathcal{L}^{-1} \left\{ \frac{1}{(s+1)^2 + 1} \right\} + \frac{2}{3} \mathcal{L}^{-1} \left\{ \frac{1}{(s+1)^2 + 4} \right\}$$

2 ile carp - böl

$$= \frac{1}{3} e^{-t} \sin t + \frac{2}{3} \cdot \frac{1}{2} e^{-t} \sin 2t$$

$$y(t) = \frac{1}{3} e^{-t} \sin t + \frac{1}{3} e^{-t} \sin 2t$$

Problemler

1) $xy'' - 2(x+1)y' + (x+2)y = xe^x$ denkleminin homojen kısmına ait bir çözüm e^x ise genel çözümünü bulunuz. $x > 0$

Homojen Denklemler: $xy'' - 2(x+1)y' + (x+2)y = 0$

$$y_1 = e^x$$

$$y_2 = xe^x$$

$$y_2' = v'e^x + ve^x = (v' + v)e^x$$

$$y_2'' = v''e^x + 2v'e^x + ve^x = e^x(v'' + 2v' + v)$$

$$v''x + 2v'x + vx - \underbrace{2(x+1)}_{-2x-2}(v' + v) + (x+2)v = 0$$

$$v''x + 2v'x + vx - 2xv' - 2xv - 2v' - 2v + xv + 2v = 0$$

$$v''x - 2v' = 0$$

$$\boxed{v' = w} \quad w'x - 2w = 0$$

$$v'' = w' \quad \frac{dw}{dx}x - 2w = 0$$

$$\int \frac{dw}{w} = \int \frac{2}{x} dx$$

$$\ln w = 2 \ln x \quad w = x^2 = v' \rightarrow \boxed{v = x^3}$$

$$y_2 = ve^x \rightarrow y_2 = x^3 e^x$$

$$y_1 \quad y_2$$

$$\begin{vmatrix} e^x & x^3 e^x \\ e^x & 3x^2 e^x + x^3 e^x \end{vmatrix} = \begin{vmatrix} 1 & x^3 \\ 1 & 3x^2 + x^3 \end{vmatrix} = 3x^2 + x^3 - x^3 = 3x^2 \neq 0$$

y_1 ve y_2 linear bağımsız

$$\boxed{y_H = c_1 e^x + c_2 x^3 e^x}$$

$$\boxed{y_P = v_1(x)e^x + v_2(x)x^3 e^x}$$

$$\begin{cases} v_1'(x)e^x + v_2'(x)x^3 e^x = 0 \\ v_1'(x)e^x + v_2'(x)(3x^2 + x^3)e^x = xe^x \end{cases}$$

$$v_1'(x) + v_2'(x)x^3 = 0$$

$$v_1'(x) + v_2'(x)(3x^2 + x^3) = x$$

Ansatz

$$v_1'(x) \Rightarrow \frac{\begin{vmatrix} 0 & x^3 \\ x & 3x^2+x^2 \end{vmatrix}}{\begin{vmatrix} 1 & x^3 \\ 1 & 3x^2+x^2 \end{vmatrix}} \Rightarrow \frac{-x^4}{3x^2+x^2-x^2} = \frac{-x^4}{3x^2} = -\frac{x^2}{3}$$

$$v_1(x) = \int -\frac{x^2}{3} dx = -\frac{x^3}{9}$$

$$v_2'(x) = \frac{\begin{vmatrix} 1 & 0 \\ 1 & x \end{vmatrix}}{3x^2} \Rightarrow \frac{x}{3x^2} = \frac{1}{3x}$$

$$v_2(x) = \int \frac{dx}{3x} = \frac{1}{3} \ln x$$

$$y_p = -\frac{x^3}{9}e^x + \frac{x^1}{3} \ln x \cdot e^x$$

Genel çözüm $y = y_H + y_p = ?$

2) $\sin^2 x y'' - \sin x y' + (\cos^2 x + 1)y = \sin x$ denli. homojen kısmının doğrusal bağımsız çözümleri $\sin x$ ve $x \sin x$ ise genel çözümü bulunuz.

homojen denklem $\rightarrow y_H = c_1 \sin x + c_2 x \sin x //$

$$y_p = v_1(x) \sin x + v_2(x) x \sin x$$

$$\begin{cases} v_1'(x) \sin x + v_2'(x) x \sin x = 0 \\ v_1'(x) \cos x + v_2'(x) (\sin x + x \cos x) = \sin x \end{cases}$$

$$v_1'(x) = \frac{\begin{vmatrix} 0 & x \sin x \\ \sin x & \sin x + x \cos x \end{vmatrix}}{\begin{vmatrix} \sin x & x \sin x \\ \cos x & \sin x + x \cos x \end{vmatrix}} \Rightarrow \frac{-x \sin^2 x}{\sin^2 x + x \sin x \cos x - x \sin x \cos x} \rightarrow -x$$

$$v_1(x) = \int -x dx = -\frac{x^2}{2} //$$

$$v_2'(x) = \frac{\begin{vmatrix} \sin x & 0 \\ \cos x & \sin x \end{vmatrix}}{\sin^2 x} \Rightarrow \frac{\sin^2 x}{\sin^2 x} \rightarrow 1$$

$$v_2(x) = x$$

$$y_p = -\frac{x^2}{2} \sin x + x^1 \sin x = \frac{x^1}{2} \sin x$$

Genel Çözüm: $y = y_H + y_p \quad y = c_1 \sin x + c_2 x \sin x + \frac{x^1}{2} \sin x$

3) $x^2 y'' - 4xy' + 6y = 6x - 6$ denk. Gözümler .

$(D(D-1) - 4D - 6)y = 6e^x - 6$ $x=e^t$ dönüşümü uygulandı
(Euler denk)

$(D^2 - 5D + 6)y = 0$

$m^2 - 5m + 6 = 0 \quad m_1 = 2, m_2 = 3$

$y_h = C_1 e^{2t} + C_2 e^{3t}$

$S = \{e^t, 1\}$

$y_p = Ae^t + B$

$y_p' = Ae^t$

$y_p'' = Ae^t$

$Ae^t - 5Ae^t + 6Ae^t + 6B = 6e^t - 6$

$6B = -6 \quad B = -1$

$2A = 6 \quad A = 3$

$y = C_1 e^{2t} + C_2 e^{3t} - 1$

$y = C_1 x^2 + C_2 e^{3x} - 2x - 1$

4) $(D-3)x + 2(D+2)y = 2\sin t$

$2(D+1)x + (D-1)y = \cos t$

sistemini "Gözümler".

$x = f(t)$
 $y = g(t)$

$$x = \frac{\begin{vmatrix} 2\sin t & 2(D+2) \\ \cos t & (D-1) \end{vmatrix}}{\begin{vmatrix} (D-3) & 2(D+2) \\ 2(D+1) & (D-1) \end{vmatrix}} = \frac{2(D-1)\sin t - 2(D+2)\cos t}{(D-3)(D-1) - 4(D+2)(D+1)} = \frac{2\cos t - 2\sin t + 2\sin t - 4\cos t}{D^2 - 4D - 3 - 4(D^2 + 3D + 2)}$$

$\Rightarrow \frac{-2\cos t}{-3D^2 - 16D - 5} = \frac{2\cos t}{3D^2 + 16D + 5}$

$(3D^2 + 16D + 5)x = 2\cos t$

$(3D^2 + 16D + 5) = 0 \Rightarrow \text{Homojen Denklem}$

$3m^2 + 16m + 5 = 0$

$(3m+1)(m+5) = 0$

$m_1 \neq -1/3 \quad m_2 \neq -5$

$x_h = C_1 e^{-5t} + C_2 e^{-1/3}$

$u = \cos t \quad S = \{\cos t, \sin t\}$

$$x_p = A \cos t + B \sin t$$

$$x_p' = -A \sin t + B \cos t$$

$$x_p'' = -A \cos t - B \sin t$$

$$-3A \cos t - 3B \sin t - 16A \sin t + 16B \cos t + 5A \cos t + 5B \sin t = 2 \cos t$$

$$-2A + 16B = 2$$

$$-16A + 2B = 0$$

$$B = 8/65 \quad A = 11/65$$

$$x = c_1 e^{-5t} + c_2 e^{-t/3} + \frac{1}{65} \cos t + \frac{8}{65} \sin t$$

$$y = \frac{\begin{vmatrix} (D-3) & 2 \sin t \\ 2(D+1) & 3 \cos t \end{vmatrix}}{\begin{vmatrix} (D-3) & 2(D+1) \\ 2(D+1) & (D-1) \end{vmatrix}}} \Rightarrow \frac{(D-3) \cos t - 4(D+1) \sin t}{-3D^2 - 16D - 5} \Rightarrow \frac{-\sin t - 3 \cos t - 4 \cos t - 4 \sin t}{-3D^2 - 16D - 5}$$

$$\Rightarrow \frac{5 \sin t + 7 \cos t}{3D^2 + 16D + 5}$$

$$(3D^2 + 16D + 5)y = 5 \sin t + 7 \cos t$$

$$(3D^2 + 16D + 5)y = 0 \quad y_h = c_1 e^{-5t} + c_2 e^{-t/3}$$

$$u_1 = \sin t \quad u_2 = \cos t \quad S = \{ \sin t, \cos t \}$$

$$y_p = A \sin t + B \cos t$$

$$y_p' = A \cos t - B \sin t$$

$$y_p'' = -A \sin t - B \cos t$$

$$5 = -3A - 16B + 5A$$

$$7 = -3B + 16A + 5B$$

$$A = 61/130 \quad B = 61/130$$